

AuthGuard-7702: Coverage-Audited Delegation-Context Analysis for Pre-Authorization Screening

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PRE-LABEL DRAFT—NOT FOR SUBMISSION.

Final human-result macros are intentionally unresolved.

Abstract—EIP-7702 lets an externally owned account execute delegate-contract code in the account’s storage, balance, and identity context. This creates a security decision before authorization, when behavioral history, reputation, and verified source may be unavailable. We present AuthGuard-7702, a bytecode-only warning and triage study built around a Delegation-Context Risk Graph (DCRG). DCRG connects reachable sensitive capabilities to recognized guard evidence and retains unresolved control flow as an explicit coverage state. A bounded extractor combines conservative state widening with jump-fenced Solidity-metadata recognition. On 1,665 unique runtimes, these corrections increase complete bounded-analysis coverage from 31.1% to 63.8% without converting a previously complete runtime to partial coverage, while inherited-label AUPRC remains statistically unchanged. We preregister a single representation endpoint on a score-blind post-cutoff sample, retrain after conservative research-family holds, and separately evaluate three new legitimate projects. Final untouched-label performance will be inserted only from the locked one-shot evaluation. The interface emits `WARN`, `NO_MODEL_WARNING`, or `DEFER`; it does not infer intent or certify authorization safety.

Index Terms—EIP-7702, Ethereum, smart-account security, bytecode analysis, selective prediction

I. INTRODUCTION

EIP-7702 changes the Ethereum account model by allowing an externally owned account (EOA) to install a delegation indicator that resolves to contract code [1]. The referenced code then executes in the EOA’s context. This enables batching, sponsorship, and scoped authorization, but a deficient delegate can expose the authorizing account to replay, unexpected calls, initialization takeover, storage collision, or broad asset control [1].

Existing EIP-7702 security studies establish that the threat is real. Huang et al. combine large-scale historical transaction filtering, Gigahorse decompilation, and cross-contract rules to identify EOA-targeted, contract-targeted, and composite attacks [2]. Qi et al. study persistent delegation phishing, activation pathways, and ecosystem concentration [3]. We ask a narrower operational question: what can be learned from a proposed delegate’s runtime bytecode *before* authorization, without requiring behavioral history?

This decision boundary creates two methodological traps. First, the inherited malicious class was constructed using static-analysis behavior, so a bytecode model can appear strong by reconstructing the labeling rule. Second, exact-runtime

separation does not prevent project or deployer dependence. Our protocol therefore separates development labels from an untouched post-cutoff human study, applies deliberately overinclusive research-family holds, freezes predictions before annotation, and reports incomplete analysis as deferral.

This study targets three contributions:

- a leakage-reduced, temporally separated evaluation resource for bytecode-only EIP-7702 screening, retaining indeterminate and non-screenable outcomes;
- a coverage-audited, guard-aware Delegation-Context representation that relates reachable sensitive capabilities to authorization evidence; and
- a warning/no-model-warning/defer evaluation over post-cutoff signer/delegate pairs, conservative family holds, strong learned and classical baselines, and separately frozen legitimate-project controls.

These are bounded claims. DCRG is represented by aggregate features consumed by XGBoost; it is not a learned graph neural network. `NO_MODEL_WARNING` is not “safe.” Anonymous linkage clusters are retraining controls, not verified project attribution.

II. BACKGROUND AND RELATED WORK

A. EIP-7702 security

An authorization tuple identifies a chain, delegate address, nonce, and EOA signature. A valid tuple writes a delegation indicator to the EOA’s code field; code-executing operations resolve the referenced delegate [1]. The specification explicitly warns that delegates should bind replay, value, gas, target, and calldata, and that initialization and storage migration are security-critical.

Huang et al. are the closest direct comparator [2]. Their transaction analyzer narrows historical activity, after which Gigahorse and rule matching identify attack semantics. AuthGuard-7702 does not replace that richer forensic setting; it studies score-blind screening at an earlier decision with bytecode as the required input. Qi et al. provide a complementary measurement and phishing analysis [3]. Accordingly, we make no claim to the first EIP-7702 detector, dataset, attack taxonomy, or connection to ERC-4337.

B. Bytecode analysis and selective prediction

Opcode models, CFG/data-flow analysis, program graphs, and cross-contract analysis are established smart-contract-security techniques [4], [5]. PhishingHook compares 16 his-

togram, vision, language, and vulnerability-model techniques for generic phishing-contract classification from EVM bytecode [6]. It therefore rules out first bytecode-phishing and first opcode-model-comparison claims. Because its labels and population differ from EIP-7702 delegate review, we do not compare published accuracy across datasets; our histogram+n-gram and learned sequence baselines are retrained under the same holds and labels as DCRG. Classification with a reject option is likewise established [7]. Our contribution is not a generic graph or rejection algorithm; it is the EIP-7702-specific representation, coverage contract, and provenance-aware evaluation at authorization time.

Full static analyzers such as Gigahorse and Securify recover richer program and compliance semantics [4], [8]; AuthGuard does not claim to replace them. Its bounded extractor exists to expose a compact, locally reproducible evidence contract and an explicit escalation path. Security-ML evaluation is also vulnerable to spatial and temporal dependence: TESSER-ACT shows how such biases inflate malware-classification estimates [9]. We apply the same principle to exact runtimes, signer/deployer clusters, research families, and time, while acknowledging that the resulting controls cannot prove independence.

III. TASK AND THREAT MODEL

The input is the deployed runtime bytecode of a proposed delegate. When available in the real-context study, the authorizing EOA recovered from the authorization signature is neutral context. A positive judgment (UNSAFE) requires a concrete authorization-relevant unsafe capability or condition. The bounded negative label, NO_CONCRETE_UNSAFE_BEHAVIOR_FOUND, means only that the reviewed evidence did not establish such behavior. INDETERMINATE and NOT_BYTECODE_SCREENABLE are reported and excluded from binary metrics.

The adversary may deploy delegate bytecode with sensitive operations, misleading selectors, unusual control flow, or appended bytes. We do not claim to resolve behavior dependent on unavailable state, unresolved proxies, external contracts, off-chain intent, or transaction-specific effects. Those cases should defer. Private-key compromise and UI deception independent of delegate semantics are outside scope.

Formally, an item is (B, A, C) , where B is runtime bytecode, A is an optional recovered authority, and C is extraction coverage. The learned models return a score $s \in [0, 1]$; held-out canonical validation determines seed-specific warning threshold t_k . Neither A nor post-cutoff labels are used to fit a threshold. The binary estimand is ranking performance among adjudicated UNSAFE and bounded-negative items. Indeterminate and non-screenable outcomes remain part of the population description but not the binary estimand.

The defender acts before signing an authorization and may lack verified source, transaction history, reputation, and downstream state. The adversary can choose bytecode, misleading selectors, unusual dispatch, appended bytes, and open

TABLE I
DCRG EVIDENCE SCHEMA. “UNKNOWN” REMAINS OBSERVABLE.

Entity	Recorded evidence	Boundary when unresolved
Entry	dispatcher targets, fallback, externally reachable regions	unreached target
Capability	calls, value movement, storage writes, approvals, creation, destruction	sensitive-site gap
Guard	self-call, signature, stored/fixed authority, EntryPoint, caller supplied, tx.origin	untyped/no recognized guard
Coverage	explored functions, unresolved jumps and state limits	PARTIAL

call/storage capabilities. We do not assume that opcode presence proves exploitability or that a known project proves legitimacy. The intended response is triage: a warning invites rejection or deeper inspection; deferral invites source audit, simulation, allowlisting, or transaction-aware analysis. The model does not replace those controls.

We organize the study around three questions. **RQ1** asks whether the coverage repairs improve analyzer validity without trading away visible gaps. **RQ2** asks whether typed guard evidence improves post-cutoff ranking beyond an untyped guard representation and fair bytecode baselines. **RQ3** asks how the frozen warning/deferral contract behaves on post-cutoff human judgments and separately sourced legitimate projects.

IV. DELEGATION-CONTEXT RISK GRAPH

A. Evidence representation

DCRG is an authority-relative evidence representation, not a claim of complete program semantics. It records four conceptual entities: externally reachable entry regions, sensitive capabilities, guard evidence, and coverage gaps. A relation records that a capability is reachable from an entry region and which recognized guard evidence dominates or precedes the site. Table I summarizes the fixed schema.

Capabilities include value-moving calls, externally parameterized targets or calldata, approvals/transfers, storage modification, contract creation, and self-destruction. A capability is not automatically unsafe: the purpose of the graph is to retain its authorization context. Guard types include self-call, recovered-signature evidence, stored authority, fixed-address comparison, ERC-4337 EntryPoint, caller-supplied comparison, and tx.origin. When the authorization tuple supplies signer A , a fixed-address comparison is additionally recorded as matching or differing from A ; transaction sender is not substituted for signer.

The classifier consumes a fixed 27-dimensional aggregate vector from this graph: function and coverage counts, guarded

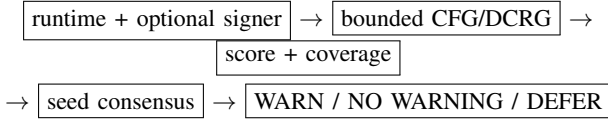


Fig. 1. The screening path. Partial analysis and seed disagreement can only warn or defer; they cannot produce `NO_MODEL_WARNING`.

and unguarded sensitive-site counts, per-guard-type counts, authority-match/mismatch counts, fallback reachability, and unreachable-site counts. Thus DCRG is a typed representation with graph-derived aggregates, not a learned message-passing architecture.

B. Bounded extraction and coverage

The extractor begins with linear disassembly, dispatcher recovery, bounded CFG construction, and a symbolic stack capped at 64 values. It explores at most 60,000 states and 96 states per program counter. After eight visits, loop-varying non-control constants are conservatively widened while valid jump destinations remain concrete. Unresolved control transfers and exploration limits are recorded as gaps rather than silently discarded.

Raw bytecode creates a second failure mode: bytes in a compiler metadata suffix can resemble `JUMPDEST`, `CALL`, or `SELFDESTRUCT`. A suspected Solidity CBOR suffix is excluded from the opcode backstop only if it decodes to a recognized shape, begins on an instruction boundary, follows a terminal predecessor, and contains no disassembled jump destination. If any fence fails, the bytes remain in analysis. These checks are correctness repairs, not a claim that metadata removal itself is novel.

The extractor emits `COMPLETE` only when all bounded obligations are discharged; otherwise it emits `PARTIAL`. No missing relation becomes evidence of absence. Figure 1 shows how this coverage state remains visible through the operating decision.

C. Scoring model and falsification variants

The primary classifier is XGBoost over the fixed aggregate vector. The capability-only variant removes guard context; the untyped variant retains guard presence but erases guard type; the actor-removed variant removes `EntryPoint` and authority-relative features. This decomposition makes the central claim falsifiable: the typed-guard contribution is supported only if full DCRG exceeds the untyped variant on the untouched endpoint. A relational GNN, local typed motifs, and fixed sequence/DCRG fusion were tested on development evidence but did not meet their preregistered or paired-improvement gates and are retained as negative results.

V. EVALUATION PROTOCOL

A. Populations and labels

The canonical development corpus contains 2,190 rows representing 1,665 unique runtimes. Its inherited target remains useful for training and implementation diagnostics but overlaps

the static evidence represented by DCRG. A separate 150-item provisional proxy informed method selection and is therefore development data.

The confirmatory population is a score-blind sample of 150 post-cutoff exact-runtime families. One predeclared development overlap is excluded from scoring. A disjoint 150-item reserve is cryptographically locked and cannot be inspected for model selection. Two independent primary reviewers label every item, and one separate adjudicator resolves each disagreement. Model scores, source labels, provisional labels, and other primary judgments are hidden.

B. Post-cutoff construction

The Ethereum collector scans a fixed checkpoint through block 24,641,536: 283,244 blocks, 517,930 type-4 transactions, and 1,233,059 authorization entries, with no block-scan RPC error. Zero-address revocations are excluded. We recover the signer from each authorization tuple rather than using transaction sender; the two differ for 574 of 708 usable signer/delegate pairs. Runtime code is read at first observed block. This is end-of-block state, so same-block changes remain a limitation.

Among 740 distinct nonzero delegate addresses, 708 pairs have historical runtime code and yield 673 exact-runtime families. Exact hash and opcode-4-gram similarity against every canonical runtime leave 564 eligible unseen families. A deterministic sampler reads no label or model column and selects at most one address per exact-runtime family. The primary and reserve use different frozen seeds and share neither item nor exact-runtime family.

C. Human judgment protocol

Two qualified primary reviewers independently inspect every primary item using the fixed taxonomy: `UNSAFE`, `NO_CONCRETE_UNSAFE_BEHAVIOR_FOUND`, `INDETERMINATE`, and `NOT_BYTECODE_SCREENABLE`. Evidence packets contain runtime, recovered signer context, decompilation and provenance aids, but recursively exclude all model scores, source-rule labels, provisional labels, and another primary judgment. Reviewers record the concrete capability, authority/caller condition, evidence boundary, and confidence. A distinct third reviewer adjudicates every primary disagreement. We report pre-adjudication marginals, raw agreement, Cohen’s κ when defined for the fixed pair, disagreement count, adjudication count, and final class counts. The evaluator refuses a partial release or a release not exactly bound to all 150 manifest IDs.

D. Leakage reduction and models

Repeated recovered authorities, EOA deployers, verified families, and overinclusive bytecode similarity links define conservative research clusters. These links produce mandatory canonical family holds; they do not prove brand identity or complete independence. Three fixed seeds train a learned sequence+dense baseline, normalized opcode histogram plus hashed 4-gram XGBoost, capability-only DCRG, untyped

TABLE II

STUDY POPULATIONS AND THEIR ALLOWED ROLE. “FROZEN” MEANS CONTENT AND IDENTIFIERS ARE HASHED BEFORE THE CORRESPONDING LABELS OR SCORES ARE OPENED.

Population	Items	Unique runtimes	Label/evidence source	Allowed role
Canonical development	2,190	1,665	inherited static source rule	training and engineering diagnostics
Gold-Test proxy	150	150 families	provisional, method-visible review	method selection only
Post-cutoff primary	150	150 families	independent dual review; pending	confirmatory, 149 score-eligible
Replication reserve	150	150 families	unopened	replacement/replication under locked rule
New legitimate projects	3	5 controls	provenance-backed operating cases	descriptive warning/defer behavior

guards, actor-removed DCRG, full DCRG, and a separately named project-balanced DCRG.

The 150 primary items form 124 dependence clusters; 19 are multi-item clusters, 45 items participate in a multi-item cluster, and the maximum size is four. Must-links use repeated recovered-authority EOAs and deployer EOAs; shared contract factories alone are not treated as ownership. A separate score-blind audit labels one item as a confirmed development overlap, assigns 148 anonymous items to conservative non-attribution clusters, and predeclares the overlap as excluded. Similarity links generate 146 mandatory canonical-family holds. This intentionally overinclusive procedure supports “leakage-reduced,” not “leakage-free.”

After holds, each seed fits on 1,205 training and 382 validation rows. The sequence baseline uses chunk attention over opcode tokens plus a dense structural branch. The classical baseline combines a normalized 225-bin opcode histogram with 512 hashed opcode 4-grams and XGBoost. DCRG variants use the same fixed seed set. The project-balanced variant additionally assigns total training weight eight to each of eight provenance-backed legitimate development projects; it is named separately and is not substituted for the primary representation comparison. Nominal 5% FPR thresholds and temperature calibration use held-out canonical validation after all holds. Retraining freezes 21 21 checkpoints and 447 primary score rows before annotation.

The single confirmatory endpoint is mean-seed AUPRC(full DCRG) minus AUPRC(untyped guards). Its 95% interval uses 10,000 paired seed-aware dependence-cluster bootstrap replicates. Baseline comparisons, AUROC, calibration, operating metrics, coverage, and error analyses are secondary or descriptive. Prevalence is always reported beside AUPRC.

For every bootstrap replicate, one dependence-cluster multiset is drawn and reused for all models and all three seeds. Candidate-minus-baseline deltas are computed per seed and averaged within the replicate before percentile bounds are taken. This avoids both item-level pseudoreplication and the invalid practice of averaging independently computed confidence-interval endpoints. The primary typed-guard hypothesis is supported only when the full-minus-untyped 95% interval has lower bound above zero. Full-minus-sequence and full-minus-histogram comparisons receive their own intervals; no global baseline-superiority statement is inferred from the primary ablation.

E. Decision contract and external controls

A seed consensus at the validation-derived nominal 5% FPR threshold emits `WARN` when at least two of three seeds warn. Exactly one warning vote emits `DEFER`. With zero warning votes, `PARTIAL` coverage or missing authority emits `DEFER`; only complete covered evidence can emit `NO_MODEL_WARNING`. A failed development experiment showed that low-score cases could not be certified as low risk, so no low-risk tier is retained. This consensus contract is a label-blind supplemental analysis frozen while the post-cutoff annotation count was zero; it does not change the preregistered AUPRC endpoint.

The separate pre-scoring external-control protocol contains Tangem, Startale, and Rainbow. It distinguishes a new project from a new runtime and a new implementation lineage. Because all three are legitimate operating cases and the sample is small, we report only project-level decisions and never AUPRC, AUROC, significance, or a population false-positive rate.

For final operating analysis, warning recall, observed warning FPR, precision, defer rate, and the observed unsafe fraction inside `NO_MODEL_WARNING` receive Wilson intervals. The latter is a failure audit, not an estimated safety guarantee. Calibration, coverage, and an error taxonomy by guard type, proxy pattern, bytecode size, and project novelty remain descriptive and cannot trigger model revision.

VI. RESULTS

A. RQ1: Coverage repair

Complete bounded-analysis coverage increases from 517/1,665 (31.1%) to 1,063/1,665 (63.8%) unique runtimes, with no complete-to-partial regression. Mean inherited-label DCRG AUPRC changes from 0.95409 to 0.95375 (−0.00034, paired 95% CI [−0.00617, 0.00423]). Thus the repair improves analysis validity without being selected for performance.

The exact-metadata intermediate appears slightly higher because syntactically valid trailers can still contain a reachable instruction boundary or jump target. The final fences intentionally restore 21 runtimes to `PARTIAL`. Across 1,665 runtimes the final extractor reports no analysis error; widening is an over-approximation and residual unresolved transfers remain visible.

TABLE III
LABEL-FREE BOUNDED-ANALYSIS COVERAGE.

Extractor	Complete rows	Complete runtimes
Original bounded CFG	670 (30.6%)	517 (31.1%)
Metadata + widening	1,453 (66.3%)	1,084 (65.1%)
Jump-fenced final	1,416 (64.7%)	1,063 (63.8%)

TABLE IV
PRE-LABEL DECISIONS ON 149 ELIGIBLE PRIMARY ITEMS.

Model	Warn	No warning	Defer
Sequence+dense	53	57	39
Histogram+4-gram	62	49	38
Capability-only	68	48	33
Untyped guards	60	52	37
Actor-removed	62	49	38
Full DCRG	62	51	36
Project-balanced	62	49	38

B. Development diagnostics, not final evidence

The provisional 150-item proxy contains 108 binary and 42 uncertain judgments and was inspected during method development. On it, full DCRG reaches 0.93885 mean AUPRC; full-minus-untyped is +0.00427 (95% paired family CI [0.00076, 0.01023]), full-minus-sequence is +0.04615 ([0.00449, 0.09178]), and full-minus-histogram is +0.03154 ([−0.01116, 0.08209]). These diagnostics justify completing the expensive independent study, but cannot appear in the abstract or support a submission claim because the proxy influenced method selection.

C. Frozen post-cutoff execution

Post-cutoff retraining produced 21 locked checkpoints and 447 score-only prediction rows (149 items $\times$ three seeds). All source, feature, hold-plan, checkpoint, and prediction hashes passed before review was unlocked. Table IV reports the resulting label-free consensus counts. Similar counts across variants are not evidence of equivalent accuracy: the items within each outcome can differ.

D. External legitimate controls

The external principal model emits 0 `WARN`, 2 `NO_MODEL_WARNING`, and 1 `DEFER`. Tangem and Rainbow are complete no-warning cases; Startale defers under partial signer-relative coverage. Rainbow is a new project but shares a known framework lineage, while the canonical-unseen subset has one no-warning and one defer. Table V keeps these distinctions explicit. This $n = 3$ project result is descriptive and does not estimate a false-alert rate.

E. RQ2–RQ3: Untouched human endpoint

The frozen release contains **[N-UNSAFE]** `UNSAFE`, **[N-BOUNDED-NEGATIVE]** bounded-negative, and **[N-INDETERMINATE]** indeterminate/non-screenable judgments; **[N-BINARY]** enter binary metrics. Raw primary

TABLE V
PRINCIPAL DECISIONS ON NEW LEGITIMATE PROJECTS.

Project	New?	Unseen?	Cov.	Decision
Tangem	yes	yes	complete	no warning
Startale	yes	yes	partial	defer
Rainbow	yes	no	complete	no warning

TABLE VI
POST-CUTOFF MODEL RESULTS. RED ENTRIES REMAIN LOCKED UNTIL INDEPENDENT REVIEW.

Model	AUPRC	AUROC
Sequence+dense	[TBD]	[TBD]
Histogram+4-gram	[TBD]	[TBD]
Capability-only	[TBD]	[TBD]
Untyped guards	[TBD]	[TBD]
Actor-removed	[TBD]	[TBD]
Full DCRG	[TBD]	[TBD]
Project-balanced	[TBD]	[TBD]

agreement is **[RAW-AGREEMENT]** and Cohen’s κ is **[KAPPA]**. Full-minus-untyped AUPRC is **[FULL-MINUS-UNTYPED]** with 95% paired interval **[PRIMARY-CI-LOWER]**, **[PRIMARY-CI-UPPER]**. For full DCRG, warning recall is **[TBD]**, observed warning FPR is **[TBD]**, warning precision is **[TBD]**, defer rate is **[TBD]**, and the observed unsafe fraction within `NO_MODEL_WARNING` is **[TBD]**.

No method-superiority conclusion is available before the independent release. The proxy-label diagnostic is intentionally excluded from this results table because it informed method selection.

VII. DISCUSSION

A. Interpreting the result branches

The final result must be interpreted at three levels. First, the primary interval determines whether typed guard categories add measurable ranking value beyond guard presence. Second, fair sequence and classical comparisons determine whether any baseline-superiority wording is supported. Third, coverage and operating decisions determine whether the representation is useful as bounded triage rather than merely a ranking model.

If the primary interval is strictly positive, the method contribution is evidence that guard type adds ranking value under the frozen post-cutoff design. It still does not imply that every guard category helps: actor-removed and capability-only comparisons are separately reported. If the interval crosses zero, the method-superiority hypothesis is rejected for this study. The paper then centers the measurement contribution: how label construction, research-family dependence, coverage correction, and explicit deferral change the evaluation of bytecode-only screening. The branch is selected mechanically and a null result is not rewritten as equivalence.

TABLE VII
FULL-DCRG FROZEN OPERATING OUTCOMES. INTERVALS ARE WILSON
95%.

Metric	Estimate	95% interval
Warning recall	[TBD]	[[TBD], [TBD]]
Observed warning FPR	[TBD]	[[TBD], [TBD]]
Warning precision	[TBD]	[[TBD], [TBD]]
Defer rate	[TBD]	[[TBD], [TBD]]
Unsafe within no-warning	[TBD]	[[TBD], [TBD]]

TABLE VIII
METHOD PATHS TESTED BEFORE THE FINAL FREEZE.

Path	Evidence gate	Disposition
Relational GNN	futility upper bound	stopped
Typed motifs	paired gain over aggregate DCRG	null
Sequence fusion	paired gain over DCRG	supporting only
Low-risk tier	bounded risk under shift	failed; removed
First-STOP normalization	semantic preservation	failed; removed
Flood-200	one valid transformation	bounded robustness result

B. Role in a defense stack

The screening point exists after a wallet learns the delegate address but before it signs the authorization. A `WARN` can block or demand stronger confirmation. A `DEFER` indicates that source audit, state-aware simulation, cross-contract analysis, reputation, or an allowlist is needed. `NO_MODEL_WARNING` says only that the fixed three-seed criterion was not met under complete bounded analysis. This division is particularly important because the EIP specification notes that changing a delegation is security-critical and that future balance behavior cannot always be inferred statically [1].

C. Negative paths as evidence

The negative paths matter. Sequence fusion did not reliably improve DCRG; the GNN met a preregistered futility stop; typed local motifs did not add a paired gain; protocol-actor features were neutral on development evidence; and the attempted low-risk tier failed. Flood-200 evaluates one synthetic transformation only. First-STOP canonicalization was not semantics-preserving and is not a defense.

These failures narrow the claim. In particular, aggregate DCRG should not be described as a GNN; deferral should not be described as a new selective-learning algorithm; and Flood-200 should not be generalized to adversarial robustness.

VIII. LIMITATIONS AND ETHICS

The post-cutoff study is Ethereum-only and covers a bounded temporal snapshot. Human judgments are evidence-bounded rather than ground truth. The analyzer is neither

sound nor complete for the EVM. Training labels still originate from a static source rule, although confirmatory labels and predictions are separated. Conservative similarity holds may overexclude while still failing to prove all project dependence absent. The external legitimate set is only three projects, one case is partial, and one shares a known framework lineage. Latency measurements do not establish wallet integration or deployment safety.

The historical runtime query observes end-of-block state rather than transaction-index pre-state. Conservative family links can both over-hold unrelated code and miss semantic or organizational relations not visible in public provenance. Shared-authority/deployer clusters address statistical dependence but do not yield project attribution. The fixed 150-item sample may contain too few bounded-negative or positive judgments for stable operating estimates; the protocol therefore requires both classes and reports intervals rather than silently proceeding.

Training on inherited source-rule labels can teach the model to reconstruct that rule. Independent post-cutoff labels prevent direct test-label circularity but cannot erase inductive bias from training. Likewise, a legitimate control is a documented operating case, not proof that all its behavior is safe. The three-project control set is intentionally reported project by project.

We analyze public bytecode, authorization records, project documentation, and audit links. A model warning is not an accusation, and a no-warning result is not an endorsement of any named project. Reviewer identities may remain pseudonymous in the artifact, but qualifications, independence, calibration, conflicts, and the private identity mapping must be retained for research integrity.

IX. REPRODUCIBILITY AND ARTIFACT

The artifact contains the canonical benchmark, exact-runtime and family identifiers, frozen post-cutoff and reserve manifests, public-provenance worklists, DCRG graphs/features, seven model variants, 21 checkpoints, score-only predictions, dependence clusters, legitimate-control evidence, and the annotation application. SHA-256 locks bind the canonical dataset, sampling code, review manifest, dependence clusters, family-hold plan, training code, feature definitions, checkpoints, and predictions. The final evaluator verifies these identities before reading labels.

The implementation uses Python 3.12.12, `evmole` 0.9.0, `cbor2` 5.7.1, `XGBoost`, and `PyTorch`. Fixed seeds are 7702, 7703, and 7704. The full revision-v3 suite contains 318 passing tests, including source locks, family isolation, prediction cardinality, dual-review/adjudication gates, label-free evidence packets, bootstrap behavior, decision consensus, and manuscript-claim auditing. The supplied build container produces a letter-paper IEEE PDF. Final LaTeX macros are generated only from the two verified evaluation reports and record both input hashes, avoiding manual result transcription.

X. CONCLUSION

AuthGuard-7702 establishes an auditable experimental framework for bytecode-only EIP-7702 pre-authorization

screening. Whether typed guard evidence provides an independent predictive advantage remains intentionally unresolved until the preregistered human evaluation. In all outcomes, the system prioritizes warning or deeper review; it does not prove malicious intent, exploitability, or safety.

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